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**YWIT Global Cybersecurity Training**

**TOPIC: SELF REFLECTION**

## Cisco Networking Basics – Modules 1 to 11

### Summary

The Cisco Networking Basics course provided me with a clear understanding of computer networking, network communication, and the technologies used to connect devices. Throughout Modules 1 to 11, I learned how networks operate and how information is transmitted between different devices.

In Module 1, Communication in a Connected World, I studied different types of networks, data transmission, bandwidth, and throughput. I understood how devices communicate and how network speed affects data transfer.

In Module 2, Network Components, Types, and Connections, I learned about clients, servers, routers, switches, modems, and other networking devices. I also explored different network types and Internet Service Provider (ISP) connectivity options.

Module 3 focused on Wireless and Mobile Networks, where I learned about wireless communication, Wi-Fi, and how mobile devices connect to networks.

In Module 4, Build a Home Network, I explored home network components, wireless standards, and the basic process of setting up a home router. The practical activities helped me understand how to connect wireless devices and configure basic network settings.

Module 5, Communication Principles, introduced me to network protocols, communication standards, and network models. I learned that protocols establish rules that allow devices to exchange information correctly.

In Module 6, Network Media, I studied different transmission media, including copper cables, fiber-optic cables, and wireless connections, and understood their uses in network communication.

Module 7, The Access Layer, introduced Ethernet frames, encapsulation, and how devices access a network. I learned how data is prepared and transmitted across an Ethernet network.

In Module 8, The Internet Protocol, I studied the purpose and structure of IPv4 addresses and their importance in identifying devices on a network.

Module 9, IPv4 and Network Segmentation, helped me understand unicast, broadcast, and multicast communication, different IPv4 address types, and how network segmentation divides a network into smaller sections for better management and performance.

In Module 10, IPv6 Addressing Formats and Rules, I learned why IPv6 was introduced, how IPv6 addresses are structured, and how they help overcome the limitations of IPv4 address availability.

Finally, Module 11, Dynamic Addressing with DHCP, explained the difference between static and dynamic IP addressing. I learned how DHCP automatically assigns IP addresses and other network configuration information to devices, reducing the need for manual configuration.

The course also included interactive learning activities, knowledge checks, practical network simulations, and checkpoint assessments that helped strengthen my understanding of the topics.

Overall, these eleven modules improved my knowledge of networking fundamentals and helped me understand how network devices, protocols, addressing systems, and communication media work together. I gained a stronger foundation in configuring basic network connections, understanding IP addressing, and recognizing the importance of reliable and efficient network communication.

These skills will be useful for my future studies and career in information technology and cybersecurity.